MATH 127 Calculus for the Sciences

Lecture 35

November 26, 2025

Today's lecture

Last time
Sigma notation
Right Riemann sums

This time
Definition of the definite integral
Properties of definite integrals
Area between curves

Information
Monday is the last day, we are going to do a final review.

Review: definition of definite integral

Let f be continuous on $[a, b]$. We divide $[a, b]$ into n equal pieces:

$$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\Delta x \quad (\text{for each } i = 1, \ldots, n).$$

Remember how the more pieces we take, the more accurate our approximation is? When you take a limit as $n \to \infty$, we in fact get the exact area.

Definition

$$\begin{aligned}\int_a^b f(x)\,dx &= \lim_{n\to\infty} (\text{right Riemann sum with } n\text{-pieces}) \\ &= \lim_{n\to\infty} \left(\sum_{i=1}^{n} f(x_i)\,\Delta x \right)\end{aligned}$$

Integral: sum and scalar multiple rules

Sum rule

$$\int_a^b \big(f(x) + g(x)\big)\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx.$$

Scalar multiple rule

$$\int_a^b c\, f(x)\, dx = c \int_a^b f(x)\, dx.$$

These look just like...

- Sigma notation:

$$\sum_{i=1}^{n} \big(cf(i) + g(i)\big) = c\sum_{i=1}^{n} f(i) + \sum_{i=1}^{n} g(i),$$

- Limit rules:

$$\lim_{x\to a} \big(cf(x) + g(x)\big) = c \lim_{x\to a} f(x) + \lim_{x\to a} g(x).$$

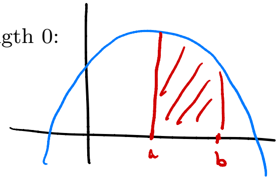
Question Why do they look so similar?

Integrals with equal or reversed bounds

Suppose we integrate f over an interval $[a, b]$.

- If the bounds are the **same**, then the interval has length 0:

$$\boxed{\int_a^a f(x)\,dx = \boxed{0}}$$



  There is no area at all.

- If you **reverse** the bounds, the sign changes:

$$\boxed{\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.}$$

$$\int_t^1 g(t)\,dt = -\int_1^t g(t)\,dt$$

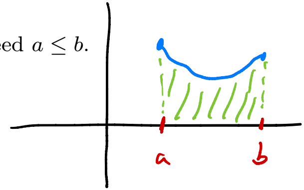
  Geometric interpretation: reversing a and b reverses the direction in which we "walk along the x–axis", so the signed area changes sign.

If you integrate a positive function, you get a positive area

Suppose f is **positive** on the interval $[a, b]$, here we need $a \leq b$.

Then

$$\int_a^b f(x)\,dx \geq 0.$$

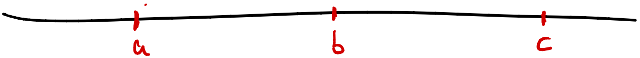


Interpretation:

- $f(x) > 0$ means the graph of $y = f(x)$ lies **above** the x–axis.
- The definite integral $\int_a^b f(x)\,dx$ is the area between the curve and the x–axis.
- If the curve stays above the axis, that signed area is positive.

Question
Could the area be equal to 0?
What can we say if $b < a$?

Additivity over two connected intervals



Integrating first from a to b, then from b to c

$\Updownarrow$ is the same as

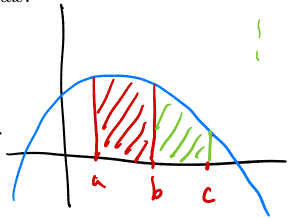
going straight from a to c

Then

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx.$$

Geometric picture:

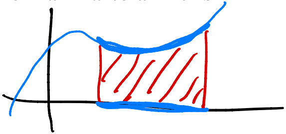
- The total area from $x = a$ to $x = c$
- equals the area from a to b **plus** the area from b to c.



Area between a curve and the x–axis

The area between the graph of $y = f(x)$ and the x–axis from $x = a$ to $x = b$ is

$$\text{Area} = \boxed{\int_a^b f(x)\,dx}$$



Or I can think of this as the area between two curves. It is the area between the curves

$$y = \boxed{f(x)} \quad \text{and} \quad y = \boxed{0}$$

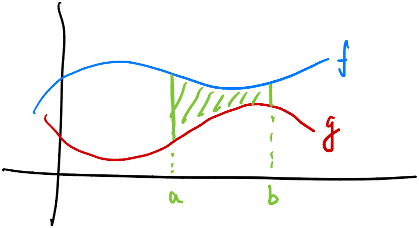
Area between two curves

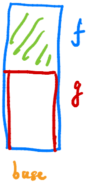
Let f and g be two functions. You get two curves from their graphs.

> The area between the curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$ is
>
> $$\text{Area} = \int_a^b \big(f(x) - g(x)\big)\,dx.$$

Idea:

- $f(x)$ gives the height of the **top** curve.
- $g(x)$ gives the height of the **bottom** curve.
- $f(x) - g(x)$ is the height of the "strip" between the curves.





$f \cdot \text{base} - g \cdot \text{base}$

$= \underbrace{(f-g)} \cdot \text{base}$

Example: area between two curves

Example Find the area between the curves

$$y = x^2 \quad \text{and} \quad y = x + 2$$

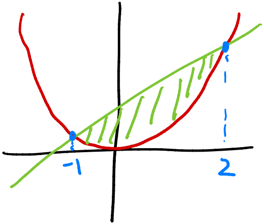
for $-1 \leq x \leq 2$.

Step 1: Graph the curves. Which one is the top curve and which one is the bottom curve?

Step 2: Set up the integral:

$$\text{Area} = \int_{-1}^{2} (\text{top} - \text{bottom})\, dx = \int_{-1}^{2} \boxed{x+2 - x^2}\, dx.$$

Step 3: Compute:



$$\int_{-1}^{2} (x+2-x^2)\,dx$$

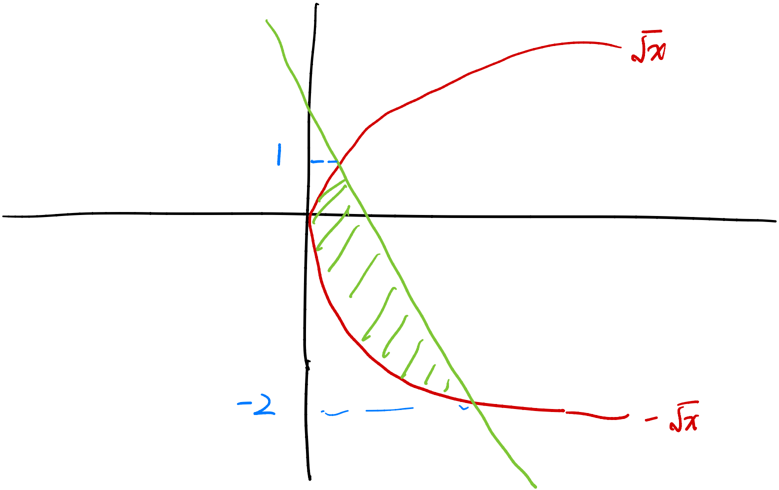
$$= \left[\frac{1}{2}x^2 + 2x - \frac{1}{3}x^3\right]_{-1}^{2}$$

$$= \left(\frac{1}{2}\cdot 4 + 2\cdot 2 - \frac{1}{3}\cdot 8\right) - \left(\frac{1}{2}\cdot 1 + 2\cdot(-1) - \frac{1}{3}(-1)\right)$$

$$= \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{9}{2}$$

Example: area between two curves

Example Make up a weird example and compute.



Example: area between two curves

Example Make up a weird example and compute.

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} \Delta x \cdot f(a+x_i)$$

approximation

differential approximation

tangent line approximation

error

$$\frac{M}{2}|x-a|^2 \qquad M=\max_{[a,x]}|f''|$$

error

$$\frac{M}{2n}|b-a|^2$$

$$M=\max_{[a,b]}|f'|$$

Example: area between two curves

Example Make up a weird example and compute.

1	2	3	4	5	6	7	8	9	10
0	2	4	8	3	6	1	1	1	1